

Behavioural repertoires tile the identifiable parameter subspaces of whole-organism nervous-system models

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Abstract

Whole-organism biophysical simulators reproduce animal behaviour from hundreds to thousands of internal parameters, yet how many of those parameters behaviour actually constrains is unknown. Here we measure the behavioural-equivalence manifold, the set of parameter configurations giving indistinguishable behaviour, across simulators spanning three phyla. Behaviour constrains only a low-dimensional stiff subspace, leaving the remaining directions free. In the nematode that subspace is the neural substrate of the animal’s four-dimensional postural repertoire, and two independently built models occupy the measured eigenworm space. Distinct behaviours constrain complementary rather than overlapping subspaces, so the identifiable dimension is the union over the behavioural repertoire and grows as behaviours are added, which we show in nematode, fly-larva and rodent models against explicit null models. The same organisation recurs in every system examined and in independent biological data. Identifiability in these models is therefore set by the richness of the behavioural repertoire.

Introduction

A central premise of mechanistic whole-organism simulators is that fitting them to behaviour recovers the biophysics that produces it. Decades of work on conductance-based models, however—from the stomatogastric ganglion (1, 2) to the broader theory of sloppy models (3, 4)—establish that many parameter sets produce indistinguishable dynamics: the map from parameters to behaviour is many-to-one (5–7). Whether, and how strongly, this degeneracy holds at the scale of a *whole-organism* nervous system—thousands of synaptic, junctional and channel parameters driving a behaving body—has not been measured. The consequences are direct: if behaviour constrains only a low-dimensional subspace, then behaviour-based calibration can recover that subspace but not the remainder, and the parameters it leaves unconstrained are candidates for the directions along which biology can vary without behavioural consequence.

Here we measure the behavioural-equivalence manifold of four published whole-organism motor simulators (BAAIWorm (8), modWorm (9), flybody (10), and NeuroMechFly v2 / FlyGym (11, 12)), which are built on physics engines for musculoskeletal control (13). To test generality, we extend the same measurement across three phyla by adding a fly-larva model, a mammalian model, a connectome-constrained *Drosophila* visual circuit and the canonical crustacean pyloric population. We do not attempt to recover the true parameters; rather, we measure the manifold of behaviourally equivalent ones, identify the constrained (stiff) subspace, and ask whether its low-dimensional geometry recurs across systems and in biology. The contribution is a measurement, together with convergent observations from biological data that use none of these models.

The components required to do this at scale are now available. Connectome-constrained whole-organism simulators rest on increasingly complete wiring diagrams (14–16) on community simulation frameworks (17–19), and on the whole-brain neural recordings they aim to reproduce (20–22); high-throughput tracking reduces behaviour to low-dimensional kinematic descriptors (23–25); and parameter inference is increasingly amortised by simulation-based methods (26–28) or performed by black-box optimisers (29, 30); behaviour-based parameter generation for whole-organism simulators has recently been demonstrated from tracking video alone (31). What none of these components settles is *how much* of a whole-organism parameter set behaviour actually determines—the classical identifiability question (32, 33) that sloppy-model theory answers for small systems (3, 4, 34) and that pyloric degeneracy answers for a single circuit (1, 2). Quantifying it at whole-organism scale is a prerequisite for behaviourally grounded digital organisms (35, 36).

Results

Behaviour converges while parameters remain dispersed

We first asked whether the manifold has positive dimension. Repeating optimisation from independent starting points against a single fixed behavioural target, the behavioural loss converged while the recovered parameter vectors remained dispersed. On the 126-dimensional FlyGym controller, 29 of 30 runs reach the behavioural target while the solution cloud occupies an effective 21.7 of 126 dimensions; on the full coupled real modWorm model (OpenWorm connectome, 279 neurons), the behavioural curvature concentrates on 2 of 7 mechanism directions (Hessian analysis below), leaving the remainder unconstrained. Behavioural convergence with persistent parameter dispersion is direct evidence that the equivalence manifold has positive dimension (Fig. 1a).

The behavioural-equivalence manifold is low-dimensional

We next quantified the dimensionality directly from the behavioural-loss Hessian at the optimum. On the 767-parameter flybody policy, a Lanczos spectrum yielded 6 stiff directions at 90% and 11 at 99% of the spectral mass, with 28 of 40 sampled directions flat to $|\lambda| \sim 10^{-11}$. On the full coupled real modWorm model, the per-mechanism behavioural Hessian had effective dimension 2 of 7 (synaptic, gap-junction and synaptic rise-time directions stiff; capacitance and gain sloppy). On the 48-dimensional FlyGym walking controller, the behavioural Hessian had effective dimension 3 (90%) / 6 (99%) of 48 and spanned 19.5 orders of magnitude. On BAAIWorm, the Gauss–Newton curvature over 20 named mechanisms had effective dimension 2 (90%) / 3 (99%) of 20. Across parameter counts of 7, 20, 48 and 767, behaviour is set by a low-dimensional stiff subspace embedded in a high-dimensional flat manifold (Fig. 1b).

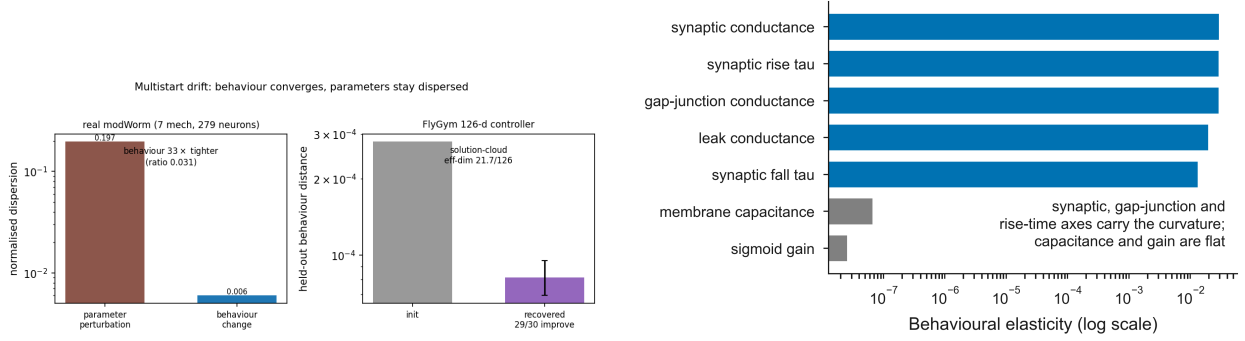


Figure 1: Behavioural convergence with parameter dispersion, and the curvature spectrum of the behavioural loss. **a** Multistart drift on two simulators. Independent optimisations started from independent points and fitted to a single fixed behavioural target converge in behaviour while the recovered parameter vectors remain dispersed. On the 126-dimensional FlyGym controller, 29 of 30 runs reach the behavioural target and the solution cloud occupies an effective 21.7 of 126 dimensions; on the full coupled real modWorm model, behaviour is approximately $30\times$ less variable across solutions than the parameters that produce it. Bars: normalised dispersion; left pair, parameter perturbation against behavioural change; right pair, held-out behavioural distance before and after recovery. Error bar: s.d. across runs. **b** Gauss–Newton behavioural curvature per biophysical mechanism on the full coupled real modWorm model (OpenWorm connectome, 279 neurons, 7 mechanisms). The synaptic and gap-junction conductances and the synaptic rise time carry the curvature; the membrane capacitance and the sigmoid gain are behaviourally flat, giving an effective dimension of 2 of 7 at both the 90% and 99% levels. Bar length: behavioural elasticity $|d \ln \text{obs} / d \ln \theta|$; colour: mechanism class. Note that behavioural convergence together with persistent parameter dispersion is direct evidence that the equivalence manifold has positive dimension, and that the curvature is carried by a small minority of directions in every system examined. Curvature is the Gauss–Newton form $H = J^\top W J$ and is therefore positive semi-definite; eigenvalues at the finite-difference floor are treated as zero. $n = 30$ independent optimisation runs (FlyGym) and $n = 10$ (BAAIWorm); dispersion is normalised per dimension.

The manifold is the neural control of eigenworm posture

The nematode behavioural manifold is not an abstract low-dimensional set but the parameter-space image of the animal’s known approximately four-dimensional postural repertoire. Real *C. elegans* posture from the OpenWorm Movement Database occupied an effective 3.79–4.14 dimensions, with four eigenworms capturing 94–96% of postural variance, reproducing the classical eigenworm decomposition (23). Two independently constructed worm models generated behaviour occupying this same empirical eigenworm space: the modWorm posture subspace aligned with the real eigenworm basis at $\cos 0.77$ and BAAIWorm at $\cos 0.68$ – 0.69 (67–72% of variance). These alignments substantially exceeded those of random four-dimensional subspaces (modWorm $p=0.001$, BAAIWorm $p=0.04$ even within a conservative ten-dimensional posture ambient; $p < 10^{-4}$ in the full space; Supplementary §6.1), and within each model it was the stiff mechanisms, not the sloppy ones, that displaced the eigenworm trajectory. In the systems examined here the effective rank of behavioural sensitivity tracks the intrinsic dimensionality of the postural repertoire, rather than the number of observables recorded: re-measuring with 7,200–14,400 posture observables in place of six summary kinematics left the effective dimension at 2–3 of 7 (Supplementary §6.2). We note that this is an empirical property of these models and behavioural losses, not a general bound: a low-

dimensional attractor can in principle constrain many parameter combinations through transients, multiple initial conditions or nonlinear temporal structure, and the intrinsic dimensionality of the behaviour need not equal the rank of the parameter-to-trajectory Jacobian. The low-dimensional behavioural-equivalence manifold is thus the neural control of the animal’s low-dimensional posture (Fig. 2a).

Distinct behaviours constrain complementary subspaces

A low-dimensional behaviour raises the question of how a whole nervous system could be identified at all. The resolution is that distinct behaviours do not constrain the *same* few directions but complementary ones. In BAAIWorm, perturbing the chemosensory parameters altered behaviour $7.15\times$ more during chemotaxis (food-directed; sensitivity 0.872) than during plain locomotion in a weak gradient (0.122), whereas perturbing the motor parameters showed the opposite contrast (0.109 versus 0.357): the axis a behaviour renders stiff is the axis its function requires, and chemotaxis and locomotion render *different* axes stiff. This is consistent with the anatomical separation of the chemosensory navigation circuit from the ventral-cord locomotor circuit (37, 38), and with models in which distinct premotor and motor elements generate the locomotor rhythm (39–41). The identifiable subspace is therefore not the few directions of any single behaviour but the union over the behavioural repertoire. Within a single worm model this union increased as behaviours were added: four distinct locomotor presets each constrained 2–4 of 7 mechanism directions with only partial overlap (mean pairwise stiff-subspace $\cos 0.54$); their behaviour-specific stiff directions spanned an effective 2.98 of 4 dimensions, each behaviour contributing on average a 0.54 component orthogonal to the others. Because a union necessarily grows whenever subspaces are not identical, we tested this against two explicit nulls. Randomly oriented directions in the same space would span 3.50 dimensions (95% interval 3.00–3.88) with mean pairwise $\cos 0.31$, whereas a single shared direction would span 1 with $\cos 1$. The observed geometry lies between the two and is significantly more aligned than random ($p=0.020$ for the span, $p=0.016$ for the pairwise cosine, 20,000 draws): the behaviours share substantial structure while each still contributes a distinct component, so the growth of the union is neither redundancy nor the trivial consequence of non-identical subspaces (Supplementary §6.4). Their union reached 5; across twelve behaviours it saturated at 4 of 7, leaving a residual 2–3 directions that no behaviour constrains, an irreducible degenerate core. The stiff and sloppy directions are distributed parameter combinations rather than single parameters (stiff-subspace participation 3.42, sloppy 4.68 of 7), so this core is a genuine biological neutral space and not an unused parameter (Fig. 2b,c).

The same geometry recurs across three phyla

This low-dimensional, complementarily tiled organisation is not specific to the nematode. We measured it also in a fly larva (larvaworld) and a mammal (Virtual Rodent (42, 43), extttm_control (44, 45)), and in each case the identifiable dimension increased as distinct behaviours were added rather than being fixed by any one behaviour. In the larva, four genuinely distinct sensory behaviours (free exploration, chemotaxis, anemotaxis and feeding) each engage a different module—olfactory, wind-sensory and feeding—so that their union (8 of 10) was approximately double the single-behaviour dimension (3–5). In the rodent, a six-behaviour repertoire drove a still-increasing identifiability curve ($3.7 \rightarrow 12$ of 38 actuators). The absolute counts are not comparable across species, as the three models are parameterised differently and the rodent is driven open-loop, but the structure is invariant: a low-dimensional stiff subspace per behaviour, complementary across behaviours, with an identifiable dimension set by behavioural richness. The crustacean pyloric pop-

ulation and a *Drosophila* visual circuit (below) extend the same stiff/sloppy geometry to a fourth and fifth system (Fig. 2d).

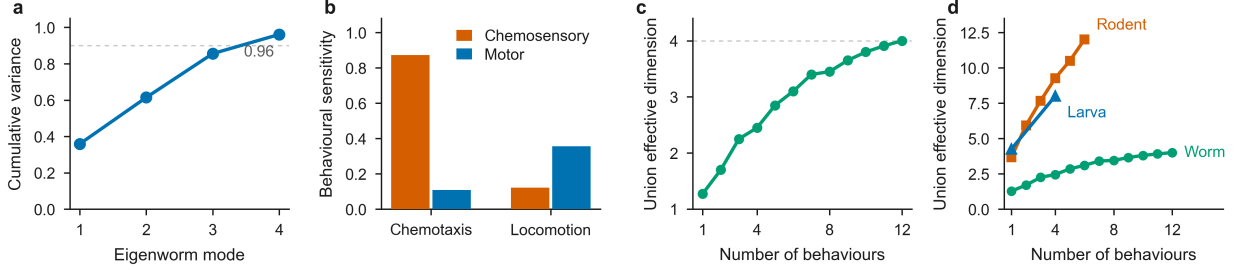


Figure 2: Identity of the low-dimensional manifold, its complementary tiling by behaviour, and the growth of the identifiable union across three phyla. **a** Cumulative postural variance of real *C. elegans* posture across eigenworm modes (OpenWorm Movement Database, $n=8432$ frames). Four modes capture 0.96 of the variance. Dashed line: the 0.90 level. Two independently constructed worm models generate behaviour occupying this measured basis, at mean principal-angle $\cos 0.77$ (modWorm) and $\cos 0.68$ (BAAIWorm). **b** Behavioural sensitivity to perturbation of the chemosensory and motor parameter classes, during food-directed chemotaxis and during plain locomotion in a weak gradient. Vermillion: chemosensory parameters; blue: motor parameters. Chemosensory parameters are stiff during chemotaxis (0.872) and sloppy during locomotion (0.122), a $7.15\times$ context shift; motor parameters show the reverse contrast (0.109 against 0.357). **c** Union effective dimension of the identifiable subspace as behaviours are added (real modWorm, twelve behaviours), saturating at 4 of 7 mechanism directions and leaving a residual core that no behaviour constrains. Dashed line: the saturation level. **d** The same increase with repertoire size in three phyla. Green: nematode (7 mechanisms); blue: fly larva (10 module parameters); vermillion: rodent (38 actuators). Note that the alignments in **a** substantially exceed those of random four-dimensional subspaces ($p=0.001$ and $p=0.04$ in a conservative ten-dimensional posture ambient, $p<10^{-4}$ in the full space; 20,000 random draws), and that absolute dimensions in **d** are not comparable across species because the three models are parameterised differently; the structure, not the count, is the invariant. Effective dimension is the number of eigenvalues reaching 99% of the spectral mass of the positive semi-definite Gauss–Newton curvature.

Two independent probes agree on the identifiable axes

Computed on the full coupled real modWorm model within a single shared prior region (per-mechanism log-multipliers, prior scale 0.15), the two geometric probes concurred on which axes are identifiable: per-mechanism Hessian stiffness and negative manifold drift were rank-correlated at Spearman $\rho=0.96$ ($p=5\times 10^{-4}$, 7 mechanisms). A third probe served as a calibration control rather than a parallel identifiability ranking: an amortised neural posterior (NPE/MAF, `sbi`) trained in the same region on $N=300$ real simulations was well-calibrated on all 7 mechanisms (per-axis simulation-based calibration passes 7/7; Kolmogorov–Smirnov $p > 0.05$). This confirms that the inference is sound but, because a calibrated posterior calibrates every axis, does not by itself yield a per-axis stiff/sloppy ordering. We therefore read the manifold’s stiff/sloppy geometry from the two probes that agree, using simulation-based calibration as a control on the inference.

Stiff and sloppy directions are biophysically interpretable

Decomposing the BAAIWorm behavioural curvature over 20 named mechanisms (Supplementary §5.2) gives a positive semi-definite Gauss–Newton spectrum (43.2, 5.84, 0.88, 0.47, remainder at the finite-difference floor) with effective dimension 2 (90%) / 3 (99%) of 20, and resolves the partition by mechanism: the stiffest axes are the passive leak, the gap-junction and synaptic weights, the motor output and several potassium conductances, whereas the fast voltage-gated calcium channels EGL-19 and UNC-2 and the calcium-activated potassium channels SLO-1/SLO-2 are the sloppiest. Behaviour is thus determined by how cells are connected and by their passive and slow properties, and is largely insensitive to the fast voltage-gated calcium conductances, consistent with the known compensation and variability of these currents in small circuits (2). A complementary probe over the full 3076-weight connectome found the connection weights collectively stiff (Supplementary §5.3), locating the degeneracy in the single-cell biophysics rather than in the connectivity. This partition is not an artefact of one model: probed identically, two independently constructed worm simulators—modWorm (coupled-ODE, 7 mechanisms) and BAAIWorm (NEURON, 20 named channels)—agree that the network-connectivity axes are stiff (stiff fraction 0.83 / 0.87) while the single-cell biophysical axes are sloppy (0.08 / 0.16), indicating that “connectivity stiff, single-cell sloppy” is reproducible across independent implementations of the same organism rather than an artefact of one model.

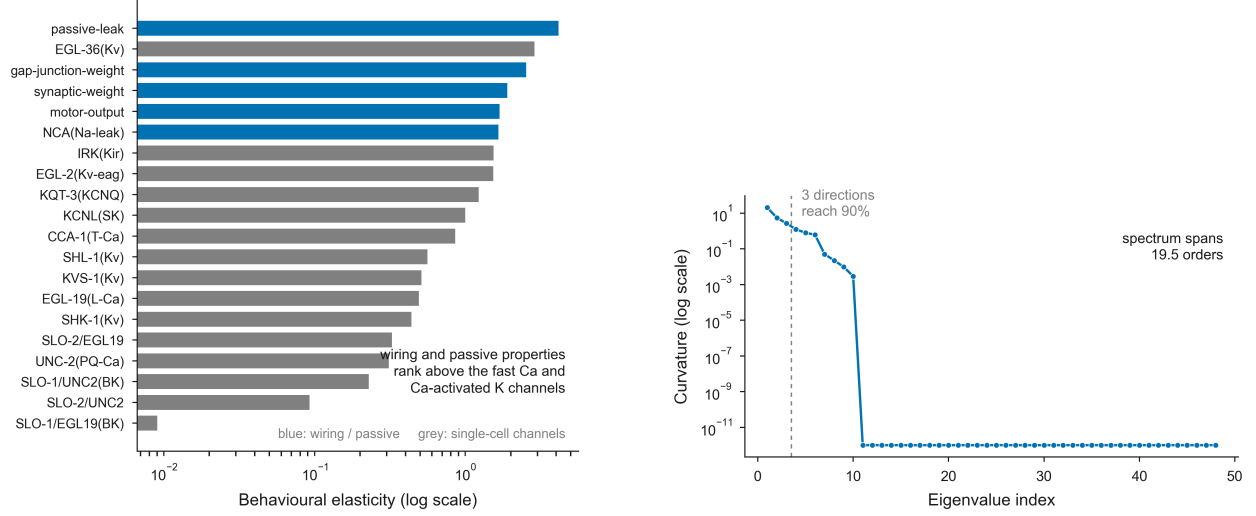


Figure 3: Stiff and sloppy structure at named-channel and higher-dimensional scale. a Behavioural elasticity of BAAIWorm over 20 named mechanisms, ordered from stiff (top) to sloppy (bottom): 16 individual ion-channel conductances, the passive leak, the synaptic and gap-junction weights, and the motor output. Bar length: $|\mathrm{d} \ln \text{obs} / \mathrm{d} \ln \theta|$ at $\delta=0.25$; colour: mechanism class. The stiffest axes are the passive leak, the gap-junction and synaptic weights, the motor output and several potassium conductances; the sloppiest are the fast voltage-gated calcium channels EGL-19 (0.49) and UNC-2 (0.31) and the calcium-activated potassium channels SLO-1 and SLO-2 (0.01–0.16). **b** Eigenspectrum of the behavioural Gauss–Newton curvature of the 48-dimensional FlyGym walking controller, sorted in decreasing order. Leading eigenvalues 20.4, 5.5, 2.7, 1.2, 0.80, 0.62, remainder below 0.06, giving an effective dimension of 3 (90%) / 6 (99%) of 48 across a spectrum spanning 19.5 orders of magnitude. Note that behaviour is set by how cells are connected and by their passive and slow properties, and is largely insensitive to the fast voltage-gated calcium conductances, which are the currents known to be homeostatically compensated in small circuits; and that a controller of substantially higher parameter dimension is still governed by a subspace of the same size. Curvature is the Gauss–Newton form $H = J^\top W J$, computed by central finite differences over the six behavioural observables and therefore positive semi-definite.

Convergent observations in independent biological data

Four independent datasets, none of which involves our forward models, exhibit low-dimensional or graded stiff/sloppy structure of the same kind (Fig. 4). These are convergent observations rather than a direct validation: they compare different levels of description—model-parameter curvature, neuronal phenotype variability, protein-sequence conservation and cross-individual behavioural variation—and so establish qualitative consistency rather than a correspondence between individual eigenvectors. (1) *Pyloric degeneracy*. Applying our Hessian probe to the Prinz–Marder population of conductance-based models that produce the same triphasic pyloric rhythm (1) (rhythm-validity gated; valid rate 0.56%) yielded a 31-parameter eigenspectrum spanning approximately ten orders of magnitude with effective dimension 2.13/31—only 2–6 stiff directions, with stiffness on the leak and inhibitory-synapse conductances and sloppiness on the fast sodium conductances, the currents known to be compensated in this circuit. This canonical degeneracy dataset exhibits the same low-dimensional stiff/sloppy partition as the simulators. (2) *Neuronal property variability*. Across $n=1914$ Allen cortical neurons, electrophysiological properties span a stiff/sloppy gradient (coefficient of variation 0.08–1.85: resting potential stiff, adaptation and firing

rate sloppy) on a property-space manifold of participation dimension 4.78/11. (3) *Gene conservation*. Across 40 *C. elegans* ion-channel and synaptic genes, ortholog identity across *Caenorhabditis* ranges from 77.9 to 99.9%; the gap-junction gene *unc-9* is the single most conserved (99.9%), paralleling the stiffness of the gap-junction axis in (1). (4) *Behavioural variation*. Across $n=61$ isogenic N2 worms (683 Tierpsy features), cross-individual behaviour occupies a participation dimension of 9.75; genetically identical animals vary on an approximately ten-dimensional behavioural manifold, the empirical counterpart of the parameter manifold. The directions that biology conserves and those it permits to vary are ordered consistently with the models' identifiable and degenerate axes, at the level of low-dimensionality and qualitative direction.

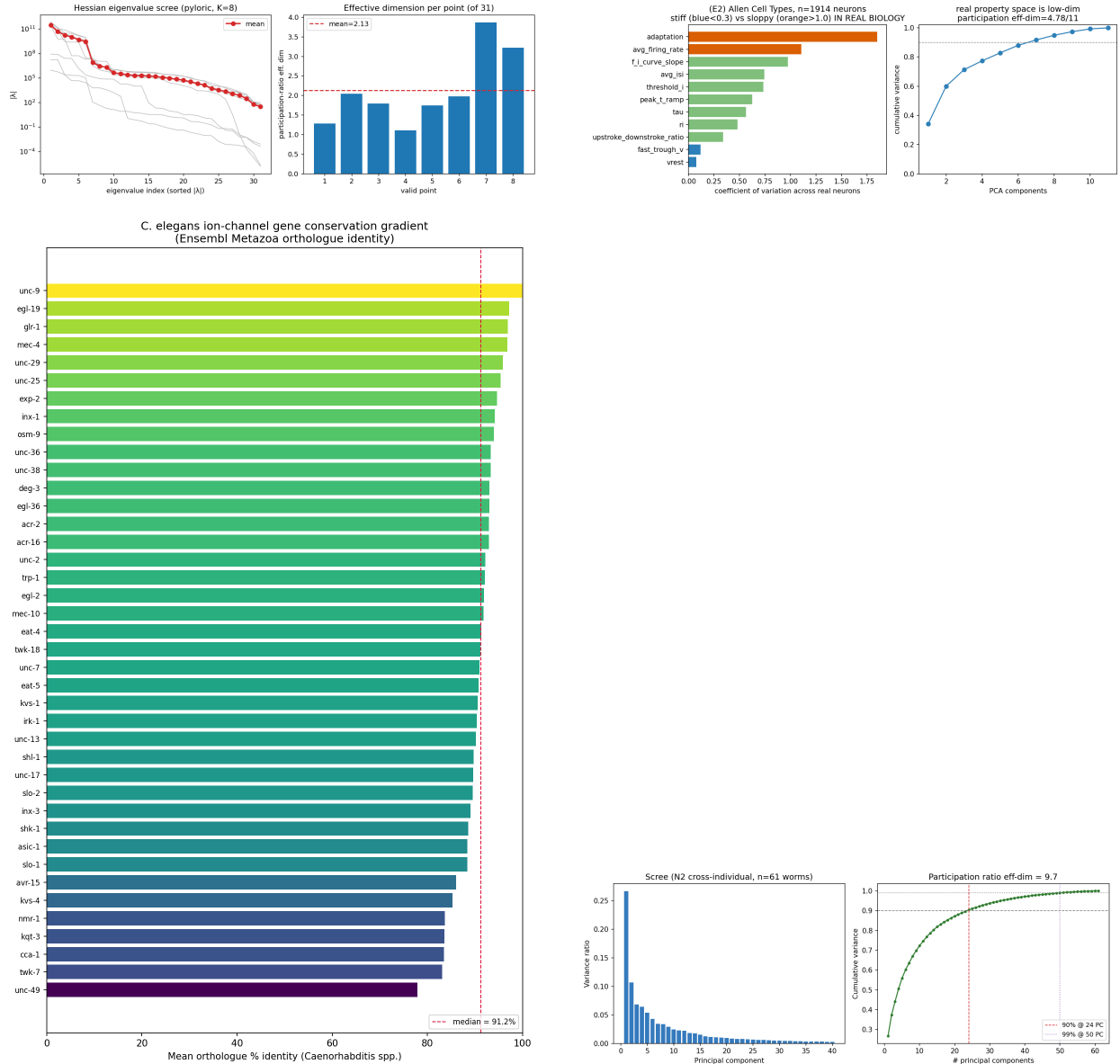


Figure 4: Low-dimensional stiff and sloppy structure in four biological datasets that use none of the simulators. **a** Prinz–Marder pyloric model population, re-gated to the canonical triphasic rhythm and probed with the identical Hessian-eigspectrum method. Effective dimension 2.13 of 31 conductances across a spectrum spanning approximately ten orders of magnitude; stiff, leak and inhibitory-synapse conductances; sloppy, fast sodium conductances. Gating: all three cells bursting, period in $[500, 2000]$ ms, strict $AB/PD < LP < PY$ phase ordering; valid rate 0.56%; $K=8$ valid points. **b** Electrophysiological property variability across Allen Cell Types cortical neurons. Coefficient of variation spans 0.08 (resting potential, stiff) to 1.85 (adaptation, sloppy); the property-space manifold has participation dimension 4.78 of 11; $n=1914$ neurons. **c** Ortholog identity of 40 *C. elegans* ion-channel and synaptic genes across *Caenorhabditis*, ranging from 77.9 to 99.9%. The gap-junction gene **unc-9** is the single most conserved at 99.9%. **d** Cross-individual behavioural manifold of isogenic N2 worms, 683 Tierpsy features, $n=61$ animals; participation dimension 9.75, with 24 principal components reaching 90% of the variance and 50 reaching 99%. Note that these are convergent observations rather than a validation: they compare different levels of description—model-parameter curvature, neuronal phenotype variability, protein-sequence conservation and cross-individual behavioural variation—and therefore establish qualitative consistency rather than a correspondence between individual eigenvectors. All four datasets are public and no simulator is used in any of them.

Discussion

We report a measurement of what behaviour can and cannot identify in whole-organism nervous-system models, applied uniformly across simulators spanning three phyla. We show that behaviour constrains only a low-dimensional stiff subspace at every parameter scale from 7 to 767, that in the nematode this subspace is the neural substrate of the animal’s four-dimensional postural repertoire, and that distinct behaviours constrain complementary rather than overlapping subspaces. The identifiable dimension is therefore the union over the behavioural repertoire, and it grows as behaviours are added.

The organising principle is that what behaviour can identify is set by how rich the behaviour is. Any single behaviour is low-dimensional and pins only a few stiff parameter combinations, but different behaviours pin the combinations their particular function requires—chemotaxis the chemosensory axes, locomotion the motor axes. Because a union grows whenever subspaces differ at all, we tested this against explicit nulls: the observed geometry is significantly more aligned than randomly oriented directions yet far from redundant, so the growth is neither trivial nor mere repetition of one shared direction.

This recasts degeneracy as a measurable resource rather than a fixed limit. A connectome-constrained simulator is identifiable up to the diversity of behaviours one is willing to elicit and measure, which makes assay diversity, rather than recording duration, the quantity an experimenter should optimise. The directions that remain after the whole repertoire has been used—the residual degenerate core—are the candidate neutral directions.

The result lifts the sloppy-models picture (3, 4, 34) and pyloric degeneracy (1, 6) from single circuits to behaving whole organisms. It also explains why behaviour-based calibration recovers a working parameter set without recovering unique biophysical values: it recovers the stiff union, not the core. The independent datasets indicate that the unconstrained directions tend also to be the more variable and less conserved ones in biology, consistent with the biophysical diversity documented within identified cell types (46).

A contrasting case bounds the scope of the low-dimensional result. Applying the same probe to a connectome-constrained *Drosophila* visual model (Supplementary §5.7) yields a much higher effective dimension (43 of 65 cell types) over a far narrower spectrum (2.9 orders of magnitude, against 19.5 for the FlyGym walking controller), while the stiff axes remain the functionally load-bearing ones, here the T4/T5 motion-detector pathway. Low dimensionality is therefore a property of the whole-body motor systems we measured, not of connectome-constrained models in general.

Is that contrast simply an artefact of how much was recorded? It is not: the worm analysis used 14,400 posture observables against 64 readouts in the visual model and still gave an effective dimension of 2–3. Two differences nevertheless remain uncontrolled—the visual model is parameterised per cell type rather than per biophysical mechanism class, and a drifting-grating response is a different kind of output from locomotion—so we treat the relation between identifiable rank and task complexity as a testable prediction rather than an established invariant.

Our approach also carries a number of drawbacks. First, the choice of behavioural observables is not unique, and a poorly chosen set could understate the identifiable rank. We therefore re-measured with 7,200–14,400 posture observables in place of six summary kinematics and recovered the same effective dimension of 2–3 of 7 (Supplementary §6.2), and we verified the estimator itself against constructed spectra of known rank (Supplementary §6.8). Second, the per-channel decomposition reaches named single conductances but still scales each conductance type uniformly across cells rather than cell by cell, and the mapping from parameter axes to biophysical classes is biophysically informed rather than learned. Two independently constructed worm models nevertheless agree on the class-level partition, which indicates that the result is not an artefact of one parame-

terisation. Third, the alignment between model stiff axes and the biological datasets is established at the level of low dimensionality and qualitative direction rather than per-eigenvector correspondence; we therefore report those datasets as convergent observations and restrict our claims to what behaviour identifies in these models.

The strongest outstanding test is prospective: predicting from the model which perturbations should produce the largest behavioural change, and testing that ordering on data not used to build it. Such a test requires perturbations resolved at the level of the coupled parameter combinations the geometry identifies rather than at the level of single genes or channels, and datasets of that kind do not yet exist for these systems. We note that defining those endpoints in advance, as we do here, carries a statistical advantage over exploratory analysis, which matters for measurements that are typically obtained from very few independent samples.

Together, these results show that behavioural identifiability is a quantity that can be measured rather than assumed, and that its extent is set by the diversity of behaviours elicited. The framework applies to any mechanistic simulator fitted to behaviour, and it turns the question "what does this model's fit actually pin down?" into one that can be answered before the fitting is done.

Methods

Manifold drift. For each simulator, N optimisations (separable CMA-ES or SPSA) from independent starts are fitted to one fixed behavioural target; we report behavioural recovery, the normalised per-dimension dispersion of the N recovered parameter vectors, and the PCA participation dimension of the solution cloud. **Behavioural curvature.** All curvature reported here is the Gauss-Newton form $H = J^T W J$, built from the finite-difference observable Jacobian $J_{ik} = \partial \text{obs}_i / \partial \theta_k$ and a diagonal positive semi-definite whitening matrix W of inverse per-observable scales. H is therefore positive semi-definite by construction and equals the Fisher information of the parameters given the behavioural observables, so its eigenvalues are non-negative and its effective dimension is the local practical-identifiability rank in the standard sense (3, 32). Eigenvalues are obtained by direct eigendecomposition (low dimension) or Lanczos iteration (high dimension). The effective dimension at level q is the smallest k such that $\sum_{i \leq k} \lambda_i \geq q \sum_i \lambda_i$ with eigenvalues sorted in decreasing order; we report $q = 0.90$ and $q = 0.99$, together with the threshold-free participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ (Supplementary §6.9). A direction counts as flat when λ falls to the finite-difference floor; residual eigenvalues of magnitude below that floor, of either sign, are numerical noise and are treated as zero. We adopt it as the primary cross-simulator metric because it is robust to the choice of observable and parameterisation, unlike a raw perturbation-sensitivity ratio, and because the manifold-drift probe provides an independent, optimisation-based cross-check that agrees with it ($\rho=0.96$). **Eigenworm identity.** Real *C. elegans* skeletons (OWMD N2) and model body postures are reduced to tangent-angle eigenmodes by PCA; we report the posture effective dimension, the top-four cumulative variance, and the principal-angle cosine between model and real eigenworm subspaces, with significance assessed against random subspaces of matched dimension. **Complementary tiling and saturation.** Per-behaviour stiff subspaces are taken as the leading Hessian eigenvectors under each behavioural target; we report pairwise stiff-subspace cosines, the union effective dimension over a behaviour set, and the saturation curve of union dimension against the number of behaviours. Context dependence (BAAIWorm) is the ratio of a parameter class's behavioural sensitivity between two behaviours (chemotaxis versus weak-gradient locomotion). The same union/saturation construction is applied across species (fly larva, larvaworld; rodent, Virtual Rodent / dm_control). **Calibration.** Per-dimension rank-based simulation-based calibration (47, 48) of an amortised neural posterior; a dimension passes when the

per-axis Kolmogorov–Smirnov test is non-significant at $\alpha=0.05$. **External datasets.** The Prinz–Marder pyloric model population re-gated to the triphasic rhythm and probed with the identical Hessian-eigenspectrum method; Allen Cell Types electrophysiology; WormBase ParaSite / Ensembl Metazoa ortholog identities; and OpenWorm Movement Database isogenic-N2 Tierpsy features. All external data are public, and no simulator is used in datasets (1)–(4). Full specifications, cohort sizes and reproduction commands are given in the Supplementary Information.

Data availability

All datasets analysed in this study are publicly available. The external biological references are: the Prinz–Marder conductance-based model population of the stomatogastric ganglion (1); the Allen Cell Types electrophysiology database (<https://celltypes.brain-map.org>); ortholog identities from WormBase ParaSite and Ensembl Metazoa (<https://parasite.wormbase.org>); and the OpenWorm Movement Database (<http://movement.openworm.org>). The whole-organism simulators are the published releases of BAAIWorm (8), modWorm (9), flybody (10), NeuroMechFly v2 / FlyGym (11), the connectome-constrained optic-lobe model of Lappalainen et al. (49), larvaworld, and the Virtual Rodent of the `dm_control` suite. All quantities reported in the main text and Supplementary Information are provided as result files, one file per analysis as named in each Supplementary section, in the deposit at <https://github.com/ChenxiHeCam/behavioural-parameter-manifold>, archived at Zenodo (<https://doi.org/10.5281/zenodo.21594746>).

Code availability

All analysis code, the per-experiment result files, and a container that reproduces every reported quantity and figure from those files are available at <https://github.com/ChenxiHeCam/behavioural-parameter-manifold> under an MIT licence, archived at Zenodo (<https://doi.org/10.5281/zenodo.21594746>). The reproduction entry point verifies each numerical claim against its source file and regenerates the figures.

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Author contributions

C.H. conceived the study, designed and performed all analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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